Iman Gholami

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PhD Candidate in Computer Science at the University of Maryland. My research combines algorithms and combinatorial optimization with LLMs and reasoning models for formal mathematics. I have shipped production-grade C++ and Python systems at scale (Balad, 30M+ installs; Bazaar, 100M+ installs) and, most recently, worked on LLM-driven code analysis at Axal (YC) and on formal verification of ZK VMs at Axiom.

SKILLS

Languages: C++ (200K+ LOC), **Python** (200K+ LOC), TypeScript, Go, Java

Core Competencies: Algorithms, Data Structures, **Problem Solving**, **Distributed Systems**, **Software Architecture**, Numerical Algorithms, Machine Learning

Frameworks, React, FastAPI, Django

Cloud: Docker, Kubernetes, AWS

DB & Tools: PostgreSQL, MongoDB, Redis, Git, CI/CD, Linux

EDUCATION

University of Maryland, College Park
PhD in Computer Science
Dec 2026

Sharif University
BSc in Computer Engineering
May 2022

AWARDS & ACHIEVEMENTS

Best Paper Award FOCS 2025

Silver Medal at IOI 2017

2nd Place at ICPC Regional (3x)

Gold Medal at INOI 2016

Ranked Competitive Programmer

PUBLICATIONS

Beyond the Library: An Agentic Framework for Autoformalizing Research Mathematics (**arXiv'26**)

Quiet Planting for k-SAT, Multiple Solutions of Arbitrary Geometry (**COLT'26**)

Bi-Criteria Metric Distortion (**ICLR'26**)

Breaking a Long-Standing Barrier: 2-ε Approximation for Steiner Forest (**FOCS'25**)

2-Approximation for Prize-Collecting Steiner Forest (**JACM'25**, **SODA'24**)

Prize-Collecting Forest with Submodular Penalties: Improved Approximation (**IPCO'25**)

Prize-Collecting Steiner Tree: A 1.79 Approximation (**STOC'24**)

EXPERIENCE

Research Assistant — University of Maryland
Jan 2023 – Present — **College Park, MD**
Designing novel approximation algorithms for Steiner Forest variants, with applications to chip manufacturing (faster performance, smaller layouts, lower energy use). Also on **DARPA mathExp** (AI for challenging mathematical problems) and **DARPA QuICC** (quantum-inspired solvers for hard SAT instances).

Research Engineer Intern — Axiom
Jun 2026 - Aug 2026 — **New York, NY**
Advancing formal verification of Zero-Knowledge Virtual Machines (ZK VMs) in Lean 4, and designed and implemented an RL evaluation system to optimize LLMs and reasoning models for automated proof generation.

Software Engineer Intern — Axal (YC W25)
May 2025 - Aug 2025 — **San Francisco, CA**
As a founding engineer, drove the software architecture and implementation of a code association system from the ground up, using LLM reasoning and static analysis to secure a modernization contract.

Software Engineer — Balad (30M+ Installs)
Nov 2018 – Dec 2022 — **Tehran, Iran**
Architected a real-time traffic prediction system (+35% ETA accuracy), overhauled the analytics stack (ClickHouse, Airflow, Spark) to cut compute costs by 30%, and refactored the core navigation service from Java to Kotlin, reducing system latency by 30%.

Software Engineer Intern — Bazaar (100M+ Installs)
Jul 2018 – Oct 2018 — **Tehran, Iran**
Built the internal wallet backing in-app transactions for 50M+ users, and debugged a critical production issue that had frozen $20,000 in user funds.

SELECTED PROJECTS

Intelligent Metadata Discovery — Metachamber
Built a lightweight metadata discovery platform that accelerated enterprise data governance by automating data catalog generation through AI-powered schema extraction and streamlined deployment.

AI Grading Assistant — Grass
Developed an AI-powered academic grading assistant that reduced TA grading effort by 90% through rubric-based scoring automation, real-time notifications, and seamless integration with existing LMS workflows.